

Radhika Gaonkar

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Research

My work in LLMs focuses on post-training, both SFT and RL, to improve reasoning capabilities, domain adaptation, and alignment for building highly capable frontier agents for M365 Copilot. I have also worked on synthetic data generation, evaluation, tool use, and translating research ideas into worldwide deployments.

Experience

Microsoft, Applied Scientist Tech Lead

Feb 2020 - Present

Microsoft 365 Copilot Research

- Led post-training and evaluation for on-policy distillation of LLMs powering [M365 Copilot](#), advancing tool use, response quality, citation accuracy, and reasoning capabilities on complex queries.
- Post-training lead for developing a novel multi-turn reasoning agent in [Copilot Tuning](#), solving planning tasks as optimization problems.
- Deployed domain-expert agents in the [Researcher](#) product with innovative mid-training strategies.
- Deployed [memory](#) feature in Copilot and pioneered an LLM-based evaluator for measuring personalization.
- Led development of an efficient LLM router framework for hallucination detection in M365 Copilot, achieving a 60% cost and a 30% latency reduction with no accuracy degradation.

User Embeddings for Office 365

- Led the development for end-to-end evaluation of user embeddings, improving the accuracy and relevance of Office 365 file recommendations.
- Deployed novel transformer-based user embeddings, boosting user engagement and file recommendation recall in Office 365 by 23%.

Smart Replies for Office 365

- Led the deployment of smart replies using LoRA-based fine-tuning of GPT-3 for enterprise customers.
- Led the fine-tuning and evaluation for deploying emoji-based smart replies in GA, increasing CTR by 3%.
- Deployed [model compression techniques](#), reducing the latency of the smart reply production model by 35%.
- Developed lexical and semantic metrics to evaluate diversity in smart replies for Outlook and Teams.

Fractal Analytics, Machine Learning Scientist

May 2016 - Dec 2017

- Deployed NLP and Computer Vision based product features for Fractal's Fortune 500 clients.
- Key contributor to Fractal's NLP and ML research division.

Technische Universität Darmstadt, Research Assistant

Jan 2014 - April 2016

- Core contributor to [UKP Lab's](#) research on argument mining and SocialNLP.
- Led research on reinforcement learning based [travel recommender systems](#).

Education

- Master of Science, Computer Science at Stony Brook University, New York 2018-2019
- Bachelor of Engineering (Honors), Computer Science at BITS Pilani, India 2010-2014

Professional Service

Industry track reviewer for NAACL 2025, NAACL 2024, ACL 2023, EMNLP 2022, NAACL 2022

Select Publications

- Permissive Information-flow analysis for LLMs - [TMLR 2025](#)
- Knowledge Infused Decoding - [ICLR 2022](#)
- Modeling Label Semantics for Predicting Emotional Reactions - [ACL 2020](#)